PRELIMINARY EXPERT RESEARCH MEMORANDUM / REVIEW COPY

PH Health Limited and PAR Health USA v. Steriscience Pte. Ltd.

The public complaint and asserted patent indicate a technically specialized ANDA dispute concerning a proposed epinephrine injection. The preliminary issue map therefore focuses on injectable-drug formulation, oxidation and light stability, excipient and container effects, analytical methods, manufacturing controls, and the relationship between the proposed generic formulation and the asserted claims. This is an issue-spotting assessment only; no opinion is offered on infringement, validity, enforceability, regulatory outcome, or commercial outcome.

Independent preliminary research based on public information.

FILED	FORUM	DOCKET	MATTER
2026-08-26	U.S. District Court for the District of New Jersey	2:26-cv-10991	Patent - Abbreviated New Drug Application (Anda)

Prepared 2026-08-27. No attorney or expert contacted. Availability, interest, compensation, and conflicts not checked.

01 / MATTER VERIFICATION AND ASSESSMENT

Public filing, likely recipient, and technical frame

CAPTION	PH Health Limited and PAR Health USA LLC v. Steriscience Pte. Ltd.
FORUM / DOCKET	U.S. District Court for the District of New Jersey / 2:26-cv-10991
FILED / TYPE	2026-08-26 / Patent - Abbreviated New Drug Application (ANDA)
PUBLIC DOCKET	Open docket

VERIFIED PUBLIC RECORD

The public CourtListener docket identifies an August 26, 2026 complaint by PH Health Limited and PAR Health USA LLC against Steriscience Pte. Ltd. in the District of New Jersey. The publicly available complaint identifies Mark M. Makhail as counsel for plaintiffs and lists his professional email on pages 12-13. The complaint concerns Steriscience's proposed generic Adrenalin (epinephrine injection) 1 mg/mL product and U.S. Patent No. 12,280,024.

LIKELY RESEARCH PURCHASER

Mark M. Makhail - Outside counsel, McCarter & English, LLP. Verified public-complaint fact: the complaint identifies Makhail as plaintiffs' counsel and lists his direct professional email on pages 12-13. That public filing role makes him an appropriate preliminary recipient for a free expert-search packet; it does not establish purchasing authority or staffing responsibility.

mmakhail@mccarter.com - publicly printed in the linked professional source; not inferred.

TECHNICAL FRAME

The public complaint and asserted patent indicate a technically specialized ANDA dispute concerning a proposed epinephrine injection. The preliminary issue map therefore focuses on injectable-drug formulation, oxidation and light stability, excipient and container effects, analytical methods, manufacturing controls, and the relationship between the proposed generic formulation and the asserted claims. This is an issue-spotting assessment only; no opinion is offered on infringement, validity, enforceability, regulatory outcome, or commercial outcome.

Secondary-source limitation: The complaint, patent, and public docket are primary public starting points, but they are not substitutes for the complete ANDA record, Paragraph IV notice, product specification, analytical data, stability program, batch records, container-closure data, prosecution history, or discovery. Open public synopsis

REPORTED ASSERTED PATENTS

U.S. Patent No. 12,280,024

Questions likely to require expert input

This issue map is a research plan, not a legal conclusion. The filed complaint, patent exhibits, intrinsic record, and discovery must control.

01 Claim-to-formulation comparison for the proposed epinephrine injection

What formulation attributes, concentrations, pH conditions, excipients, and preparation steps do the asserted claims require, and which are actually present in Steriscience's proposed 1 mg/mL epinephrine injection?

Potential evidence: Asserted claims and specification; prosecution history; ANDA product description; batch records; certificates of analysis; product and release specifications; expert formulation comparison.

02 Epinephrine degradation pathways

How do oxidation, light exposure, dissolved oxygen, trace metals, pH, and processing conditions affect epinephrine potency and degradant formation, and do the claimed and proposed formulations address the same mechanism?

Potential evidence: Forced-degradation studies; photostability and oxidative-stress data; chromatograms; degradant identification; container-headspace information; published pharmaceutical-stability literature.

03 Excipient function and formulation stability

What functional role do antioxidants, chelators, buffers, tonicity agents, and other inactive ingredients play in an injectable epinephrine formulation, and are observed stability results attributable to the claimed combination rather than ordinary formulation practice?

Potential evidence: Formulation-development reports; excipient compatibility studies; literature; formulation screening data; design-of-experiments records; product specifications.

04 Sterile manufacturing and container-closure effects

Could compounding, sterilization, filling, oxygen control, packaging, or container-closure selection materially change epinephrine stability or account for any differences between the products?

Potential evidence: Manufacturing batch records; aseptic-process validation; fill and headspace controls; extractables/leachables work; container-closure studies; storage-condition data.

Questions likely to require expert input

05 Analytical-method reliability

Do the analytical methods used to establish potency, impurities, degradants, and shelf life reliably distinguish the technical features asserted from ordinary product quality attributes?

Potential evidence: Method validation; HPLC or LC-MS methods; system-suitability and recovery data; stability protocols; raw chromatographic data; impurity reference standards.

06 ANDA bioequivalence and pharmaceutical equivalence context

What can, and what cannot, be inferred from the ANDA's pharmaceutical-equivalence and bioequivalence showing about the composition, stability, and performance of the proposed product?

Potential evidence: ANDA chemistry, manufacturing, and controls materials; bioequivalence protocols and reports; FDA correspondence; product labels; regulatory guidance.

07 Priority-date formulation baseline and validity

What did the literature, product labels, patents, and ordinary formulation practice teach before the relevant priority date about stabilizing epinephrine injection, and would the claimed feature set have been expected to deliver the reported result?

Potential evidence: Priority-date publications and patents; historical product labels; pharmacopeial materials; formulation textbooks; laboratory notebooks; contemporaneous regulatory guidance.

Candidates 1-2 for further diligence

Ranked for preliminary research priority based on subject-matter fit, relevant public work, testimony, and visible diligence burden. This is not a retention recommendation.

01 Diane J. Burgess, PhD

Professor of Pharmaceutical Sciences, University of Connecticut School of Pharmacy

TECHNICAL FIT

Direct fit for parenteral and injectable-drug formulation, drug delivery, excipient function, physical stability, and formulation development.

RELEVANT WORK

Her University of Connecticut profile identifies research in pharmaceutics and drug delivery, including formulation and dosage-form development relevant to injectable products.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

Her profile is broad across dosage forms; a party-specific screen should confirm direct epinephrine, injectable, sponsor, and prior-testimony history before considering retention.

Source 1

Draft email | d.burgess@uconn.edu | public email source

02 Patrick J. Sinko, PhD, RPh

Professor of Pharmaceutics, Rutgers Ernest Mario School of Pharmacy

TECHNICAL FIT

Strong fit for formulation science, biopharmaceutics, drug delivery, and translating drug-product attributes into performance evidence.

RELEVANT WORK

Rutgers identifies Sinko as a pharmaceutics faculty member whose work spans drug delivery and biopharmaceutics.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

His broad drug-delivery work may require supplementation by a sterile-products or epinephrine-stability specialist; prior consulting, sponsored research, and testimony require diligence.

Source 1

Draft email | sinko@pharmacy.rutgers.edu | public email source

Candidates 3-4 for further diligence

03 Robert O. Williams III, PhD

Professor of Molecular Pharmaceutics and Drug Delivery, The University of Texas at Austin

TECHNICAL FIT

Strong formulation and process-development fit for dosage-form design, excipient selection, manufacturing effects, and characterization of drug products.

RELEVANT WORK

The University of Texas expert profile identifies Williams as a pharmaceutical-sciences researcher in molecular pharmaceutics and drug delivery.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

His visible research record includes a range of delivery systems beyond sterile epinephrine injections; direct experience with solution stability and ANDA practice should be confirmed.

Source 1

Draft email | bill.williams@austin.utexas.edu | public email source

04 Stephan Schmidt, PhD

Professor of Pharmaceutics, University of Florida College of Pharmacy

TECHNICAL FIT

Useful for ANDA, bioequivalence, pharmacometrics, and connecting formulation and manufacturing attributes to drug exposure or performance evidence.

RELEVANT WORK

The University of Florida profile identifies Schmidt as a pharmaceutics faculty member working in pharmacometrics and quantitative drug development.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

His greatest value may be regulatory and quantitative-performance issues rather than oxidation chemistry; a pure stability expert may still be required.

Source 1

Draft email | sschmidt@cop.ufl.edu | public email source

Candidates 5-6 for further diligence

05 Robin A. Bogner, PhD

Professor of Pharmaceutical Sciences, University of Connecticut School of Pharmacy

TECHNICAL FIT

Direct fit for pharmaceutical physical chemistry, dosage-form stability, excipient effects, and formulation characterization.

RELEVANT WORK

The University of Connecticut profile identifies Bogner as a pharmaceutical-sciences faculty member; her scholarly work concerns physical pharmacy and dosage-form behavior.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

A current-role and availability check would be needed before any later authorized contact; pharmaceutical physical chemistry may need to be paired with a sterile-injectable manufacturing expert.

Source 1

Draft email | robin.bogner@uconn.edu | public email source

06 Steven P. Schwendeman, PhD

Professor of Pharmaceutical Sciences and Biomedical Engineering, University of Michigan

TECHNICAL FIT

Strong fit for mechanistic drug stability, formulation, degradant pathways, excipient interactions, and analytical interpretation.

RELEVANT WORK

His publicly available University of Michigan CV identifies work in pharmaceutical sciences and drug delivery, including formulation and stability-related research.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

His principal research includes complex delivery systems, so the exact relevance to aqueous epinephrine injection should be established through publications and project history; sponsor relationships require review.

Source 1

Draft email | schwende@umich.edu | public email source

Candidates 7-8 for further diligence

07 David B. Volkin, PhD

Professor of Pharmaceutical Chemistry, University of Kansas School of Pharmacy

TECHNICAL FIT

Strong fit for drug-product stability, degradation mechanisms, analytical characterization, and formulation risk assessment.

RELEVANT WORK

The University of Kansas profile identifies Volkin as a pharmaceutical-chemistry faculty member whose research addresses pharmaceutical product quality and stability.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

His public record is particularly strong in biologics and protein stability, so the fit to small-molecule epinephrine and sterile injection must be confirmed at a detailed level.

Source 1

Draft email | volkin@ku.edu | public email source

08 Thomas J. Anchordoquy, PhD

Professor of Pharmaceutical Sciences, University of Colorado Anschutz Medical Campus

TECHNICAL FIT

Useful for formulation stability, physicochemical mechanisms, excipient interactions, and experimental design for drug-product comparison.

RELEVANT WORK

The University of Colorado profile identifies Anchordoquy as a pharmaceutical-sciences faculty member with research in drug delivery and formulation.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

The public profile emphasizes delivery systems, which may be less directly focused on conventional epinephrine injection; current industry relationships and product-specific experience require diligence.

Source 1

Draft email | tom.anchordoquy@cuanschutz.edu | public email source

Candidates 9-10 for further diligence

09 Raj Suryanarayanan, PhD

Professor of Pharmaceutics, University of Minnesota College of Pharmacy

TECHNICAL FIT

Good fit for excipient behavior, physical pharmacy, phase behavior, processing, and stability questions bearing on formulation equivalence.

RELEVANT WORK

The University of Minnesota faculty profile identifies Suryanarayanan as a pharmaceutics researcher with work in physical pharmacy and pharmaceutical formulation.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

His research may be more focused on solid-state and physical-pharmacy questions than aqueous sterile products; an injectable-stability specialist may be needed for certain opinions.

Source 1

Draft email | surya001@umn.edu | public email source

10 Christian Schoneich, PhD

Professor of Pharmaceutical Chemistry, University of Kansas School of Pharmacy

TECHNICAL FIT

Strong chemistry fit for oxidation and photodegradation mechanisms, stability-indicating analytics, and interpretation of degradation evidence.

RELEVANT WORK

The University of Kansas profile identifies Schoneich as a pharmaceutical-chemistry faculty member with research in drug stability, oxidation, and degradation pathways.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, deposition, trial testimony, or admissibility decision was identified in the bounded public sources used for this preliminary record.

POINTS FOR FURTHER DILIGENCE

His best-documented work includes complex molecular and protein degradation, so a careful review should establish the precise small-molecule epinephrine overlap; past sponsorship and testimony require diligence.

Source 1

Draft email | schoneic@ku.edu | public email source

Preliminary candidates; no conflicts or availability determination

- Run party, affiliate, counsel, inventor, funder, and technology conflicts.

- Confirm testimony history, exclusions, compensation, publications, patents, and deposition performance.

- Investigate licensing, grants, consulting, commercial interests, and prior technical positions.

- Confirm role fit, availability, and interest only after counsel authorizes outreach.

ONGOING MATTER-SPECIFIC RESEARCH

Flint Witness Research works in the post-filing window, turning pleadings and public technical records into issue maps, researched candidate slates, and diligence starting points as the case develops.

Schedule a 15-minute conversation

CORE SOURCES

Public docket: https://www.courtlistener.com/docket/74704075/ph-health-limited-v-steriscience-pte-ltd/

Public professional email source: https://storage.courtlistener.com/recap/gov.uscourts.njd.606874/gov.uscourts.njd.606874.1.0.pdf

Public technical context source: https://storage.courtlistener.com/recap/gov.uscourts.njd.606874/gov.uscourts.njd.606874.1.0.pdf

LIMITATIONS

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